RESEARCH ARTICLE SUMMARY

NEUROSCIENCE

Flexible recruitment of memory-based choice representations by the human medial frontal cortex

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Journal Club in CNBI Lab

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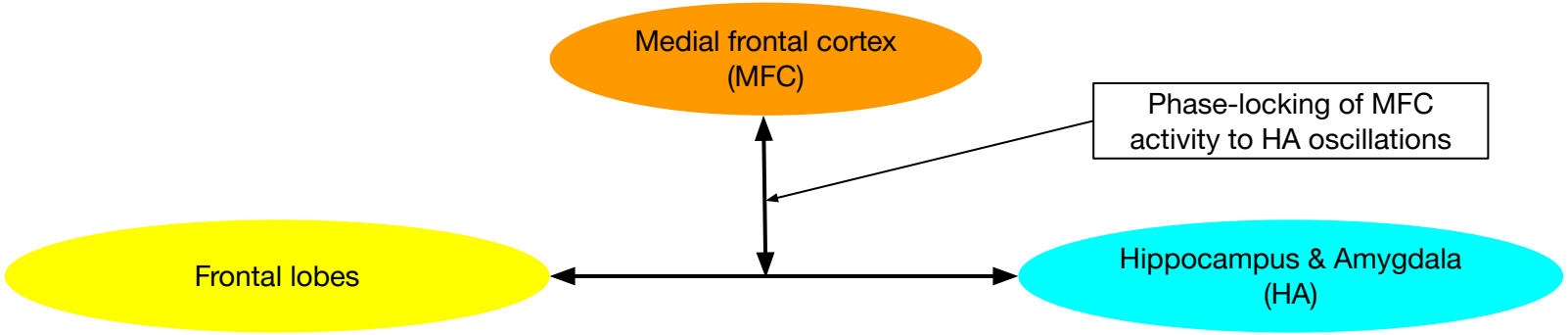
Introduction

Decision making = Stimulus representation + Memory & Context



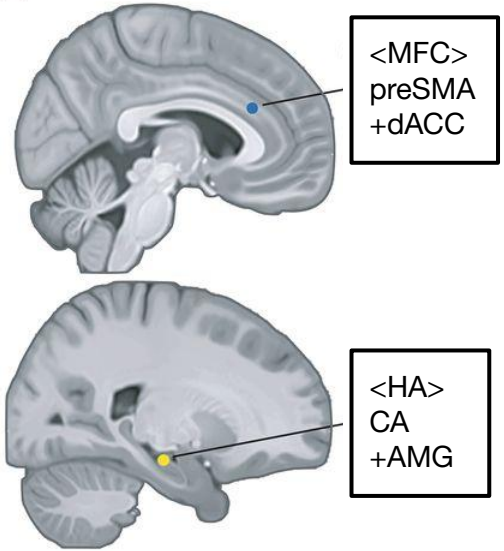
Remaining questions

1. What features of decisions and context are represented in the MFC; and
2. What functional interactions between the MFC and HA mediate dynamic memory retrieval during a task is unknown.

Introduction

Their unique approach

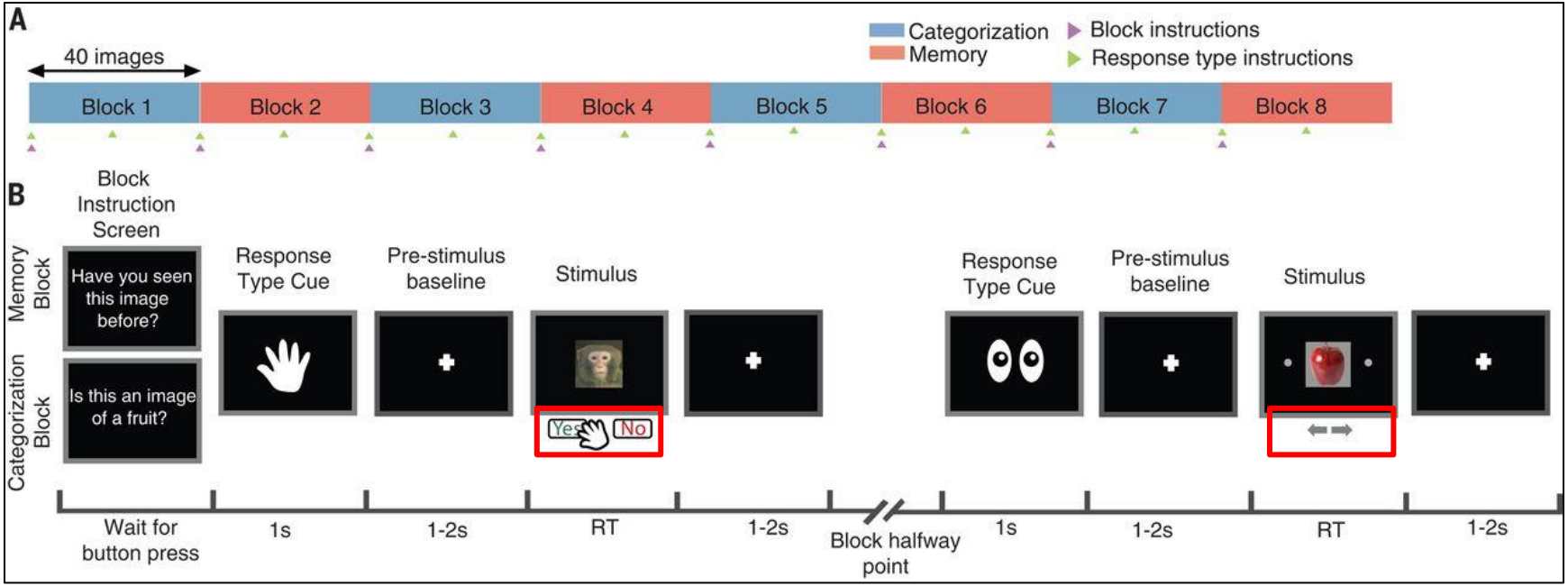
- Recorded single neurons and local field potentials in the human MFC and HA in patients implanted with depth electrodes.
- Subjects switched between recognition memory and categorization tasks.
- Compared the strength of encoding of **stimulus familiarity**, **category**, and **choices** between tasks.
- Tested cross-task generalizability of decoders.



Results

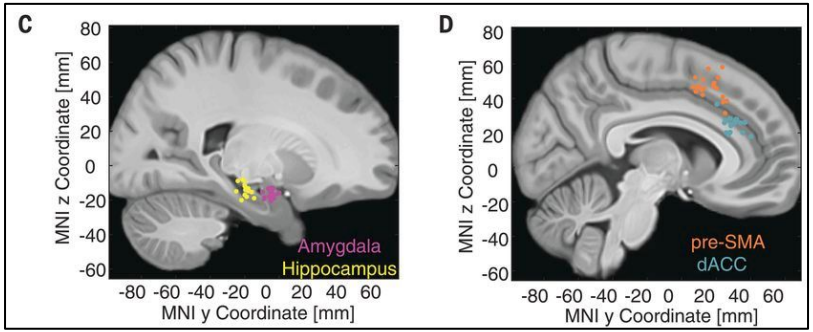
1. Cells in the MFC represented **task set** during baseline periods.
2. The strength and geometry of representations of **familiarity** were task-insensitive in the HA but not in the MFC.
3. The **visual category** of stimuli was represented more strongly during the **memory task** in both the MFC and HA.
4. **Choices** in both tasks were most strongly represented by cells in the MFC.
5. MFC cells phase-locked their activity to theta-frequency band oscillation in the HA preferentially in the **memory task**, with memory-choice cells also phase-locking in the gamma-frequency band. The strength of this interareal phase-locking was predictive of behavior.

1. Task



- Subjects were viewed a sequence of 320 images, grouped into 8 blocks of 40 images each.
- Subjects made a 'yes' / 'no' decision for each trial to indicate whether an image:
  1) belonged to a given visual category; or ⇐ Categorization task
  2) had been seen before in the task or not ⇐ Memory task

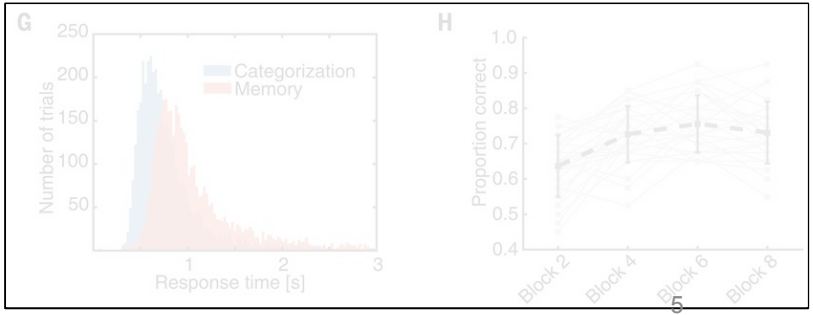
2. Electrode locations and behavior



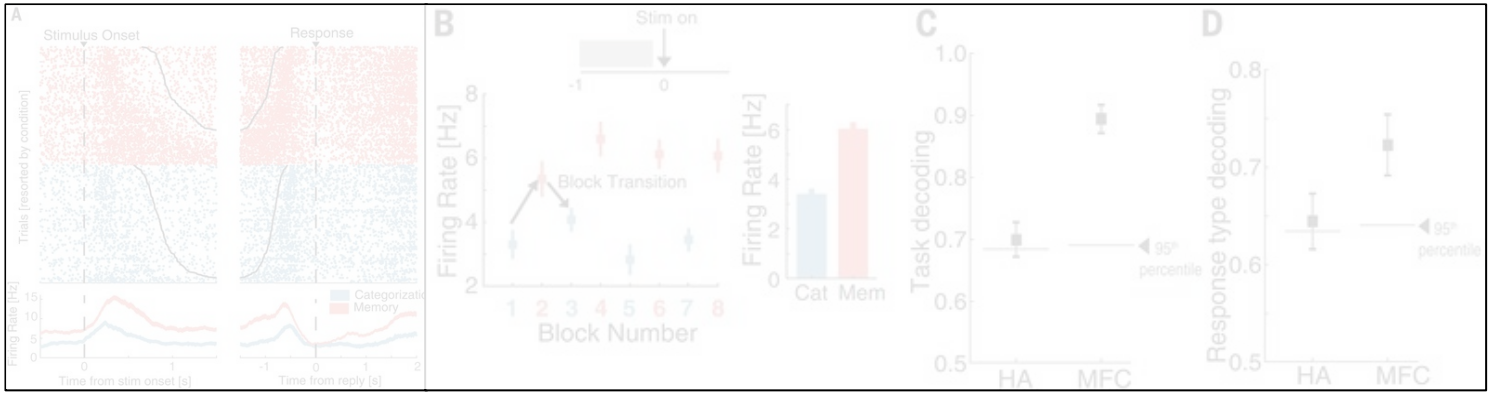
- 33 sessions in 13 subjects.
- Recordings of 1430 single neurons across four brain areas:

Area		Referred name	Neurons #
1	Anterior hippocampus (HF)	Hippocampus & amygdala (HA)	663
2	Amygdala (AMY)		
3	Dorsal anterior cingulate cortex (dACC)	Medial frontal cortex (MFC)	767
4	pre-supplementary motor area (pre-SMA)		

- (**G**) - Reaction times (RTs) were significantly longer in the memory task than in the categorization task. (1.48 s vs. 1.19 s)
- (**H**) - In the memory task, accuracy increased as a function of how many times an image had been shown.



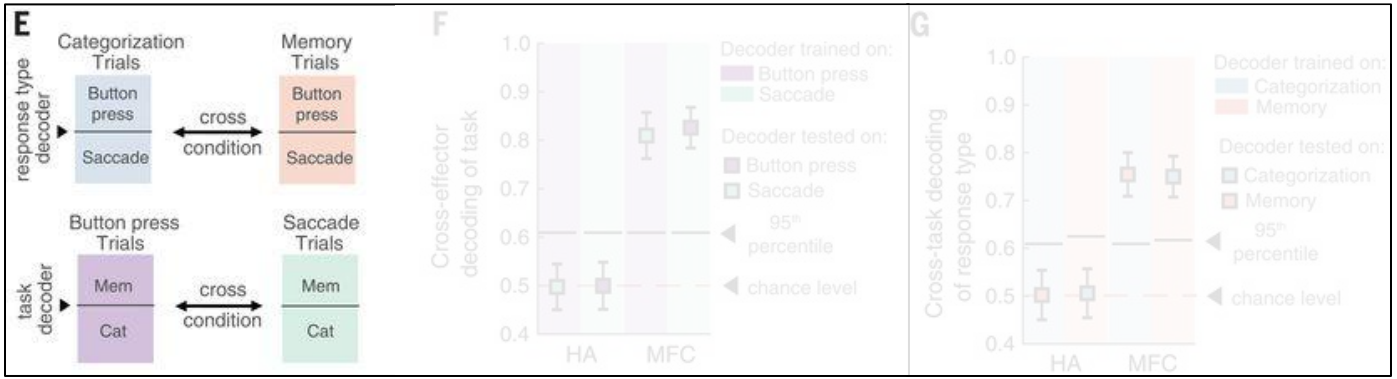
2. Task and response type representations



- (**A**,**B**) - Cells showed significant modulation of their firing rate during the baseline period as a function of task type.
- At the single-neuron level, significantly more cells were modulated by **task type** in the MFC than in the HA. (25% vs. 12%)

- (**C**) - Population decoding accuracy was significantly higher in the MFC than in the HA. (90% vs. 70%)
- (**D**) - Significantly more cells encoded **response modality** in the MFC. (14% vs. 10%)
- This signal could be decoded with higher accuracy in the MFC than HA. (72% vs. 64%)

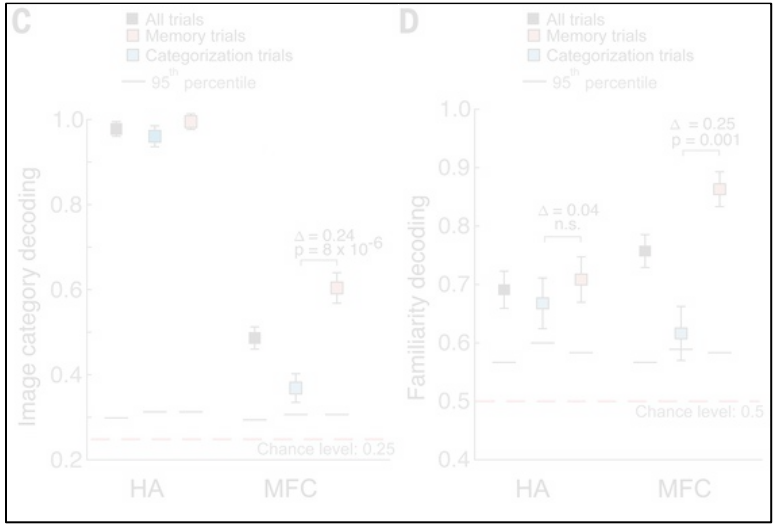
2. Task and response type representations



- (**E**) - Tested whether the two types of contextual signals were sufficiently robust to avoid interference with one another, using a cross-condition generalization decoding analysis.
  1. Trained a decoder to discriminate task type on trials where the subject was instructed to reply with a button press.
  2. Tested the performance of this decoder on trials where the subject was instructed to use saccades.

  ⇒ (**F**,**G**) - The two decoders **generalized** in the MFC but not in the HA.

3. Image category and familiarity representations



- Used decoding performance on individual trials and single-neuron analysis to assess whether the neural encoding of visual category and/or familiarity of a stimulus depended on **task** demands.
- Visually selective cells - reflect input from high-level visual cortex.
- Memory selective cells - response signals stimulus familiarity.

- (**C**) - Image category was decodable in both HA and MFC.

➔ HA - no significant difference in decoding accuracy. (96% vs. 99%)

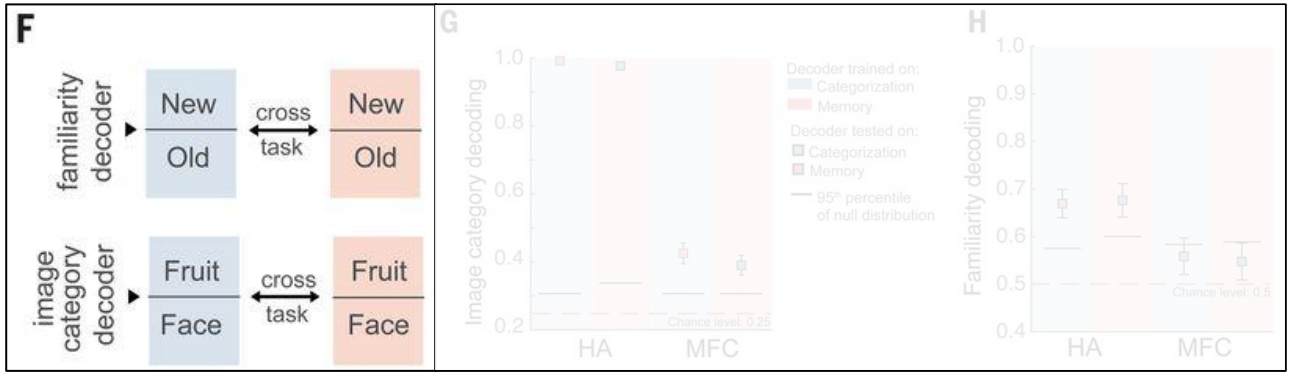
➔ MFC - higher decoding accuracy for memory tasks. (60% vs. 36%)

- (**D**) - Memory was decodable in both HA and MFC.

➔ HA - no significant difference in decoding accuracy. (67% vs. 71%)

➔ MFC - higher decoding accuracy for memory tasks. (86% vs. 61%)

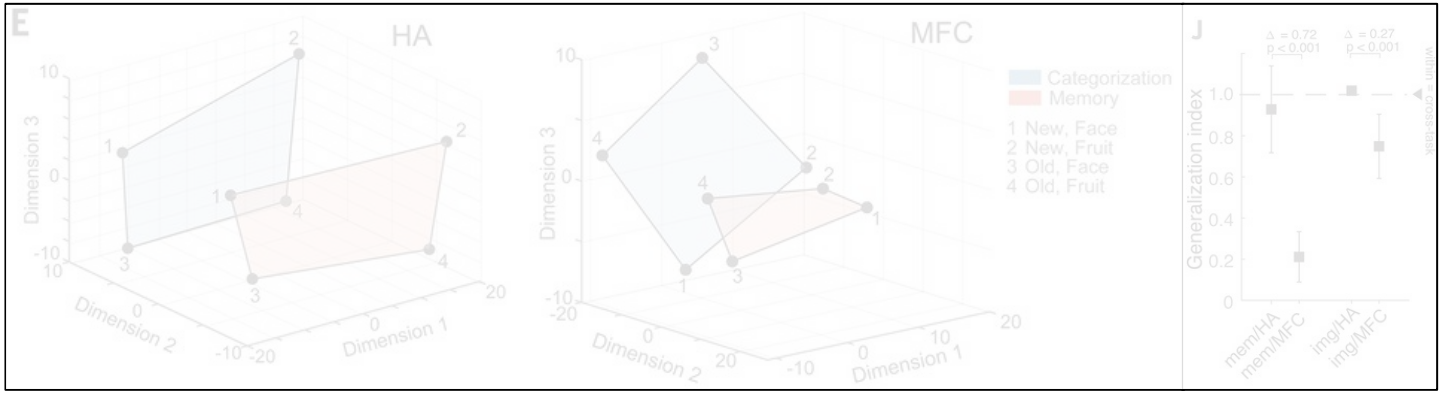
3. Image category and familiarity representations



- (**F**) - Assessed whether the decoders trained to report familiarity and the category of the stimuli in one task would generalize to the other task.
- Cross-task **generalization** performance was greater in the **HA** than MFC for both image category (**G**, **I**) and familiarity (**H**, **I**).

➔ Show that the geometry of the representations of stimulus familiarity and memory were significantly less sensitive to task demands in the HA compared with the MFC.

3. Image category and familiarity representations



- Projected the average HA and MFC population activity for all possible pairings of familiarity, image category, and task into a 3D state space using multidimensional scaling (MDS).

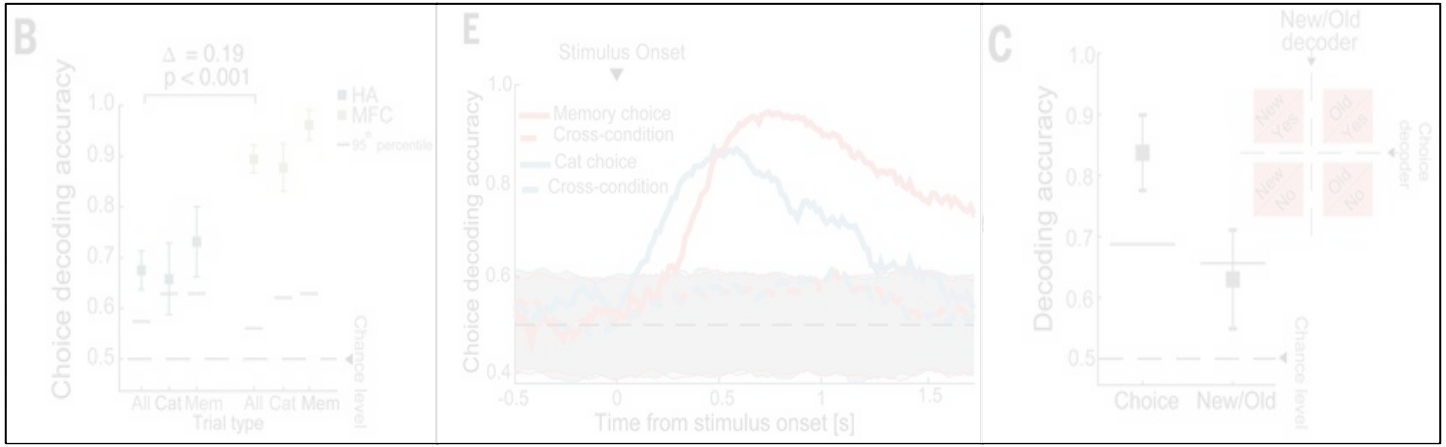
❖ MDS using Euclidean distances on z-scored spike count data in the [0.2 s, 1.2 s] window relative to image onset.

➔ HA - relative position of 'new face' with respect to 'old face' were preserved across tasks.

➔ MFC - relative positions of the four conditions were not preserved.

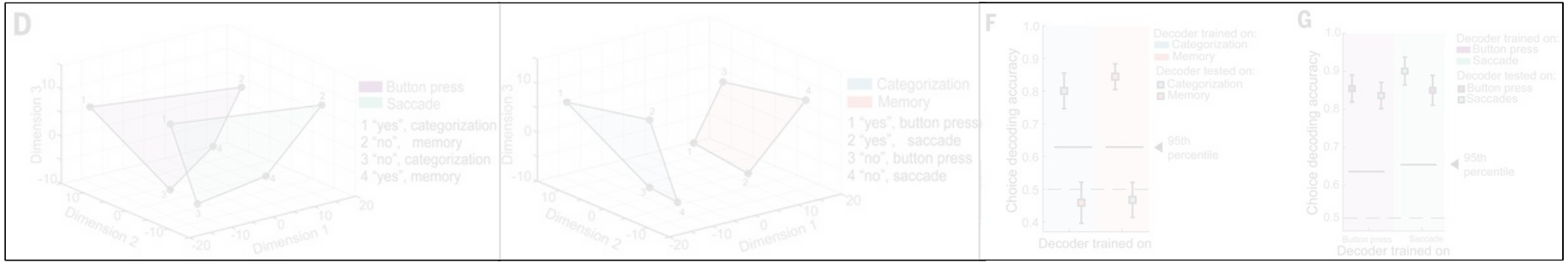
★ (**J**) - Consistent with the weaker cross-task generalization performance observed in MFC relative to HA, resulting in reduced generalization indices in the MFC compared with HA.

4. Choice representations



- (**B**) - Decoding accuracy for choices (yes/no) was highest in the MFC, with an average population decoding performance of 89% compared with 68% in the HA.
- (**E**) - Choice signal was significantly stronger in the MFC regardless of selection threshold and quantification method.
- (**C**) - The activity of MFC cells predicted choice but not the ground truth at levels significantly different from chance.

4. Choice representations



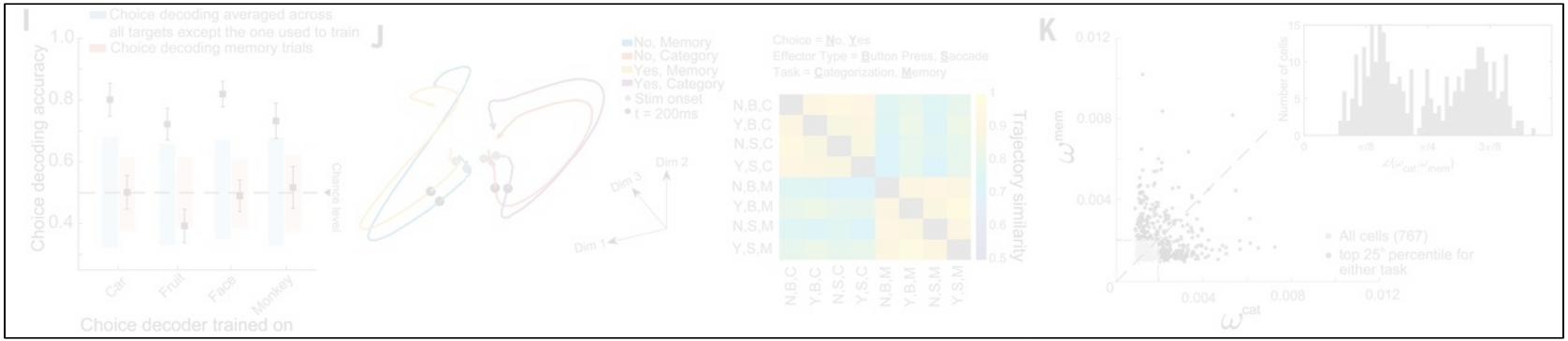
- Used multidimensional scaling to to visualize the population activity for the eight combinations of choices, task types, and response modality.

➔ Geometrical configuration indicates that **choice** decoding generalizes across response modality but not across task types.

- Computed the cross-task generalization performance of a decoder trained on choices during one task and tested on the other and performed this analysis across time and also in a single post-stimulus time bin.

➔ While the choice signal did not generalize across task types (**F**), it did generalize across response modality within the same task type (**G**).

4. Choice representations



- (**I**) - Choice decoding generalized across all subtasks in the categorization task but not the memory task.
- (**J**) - Comparing the pairwise similarity between the trajectories in **state space** within the first 500 ms after the stimulus onset revealed that patterns of dynamics in state space first cluster by task type.

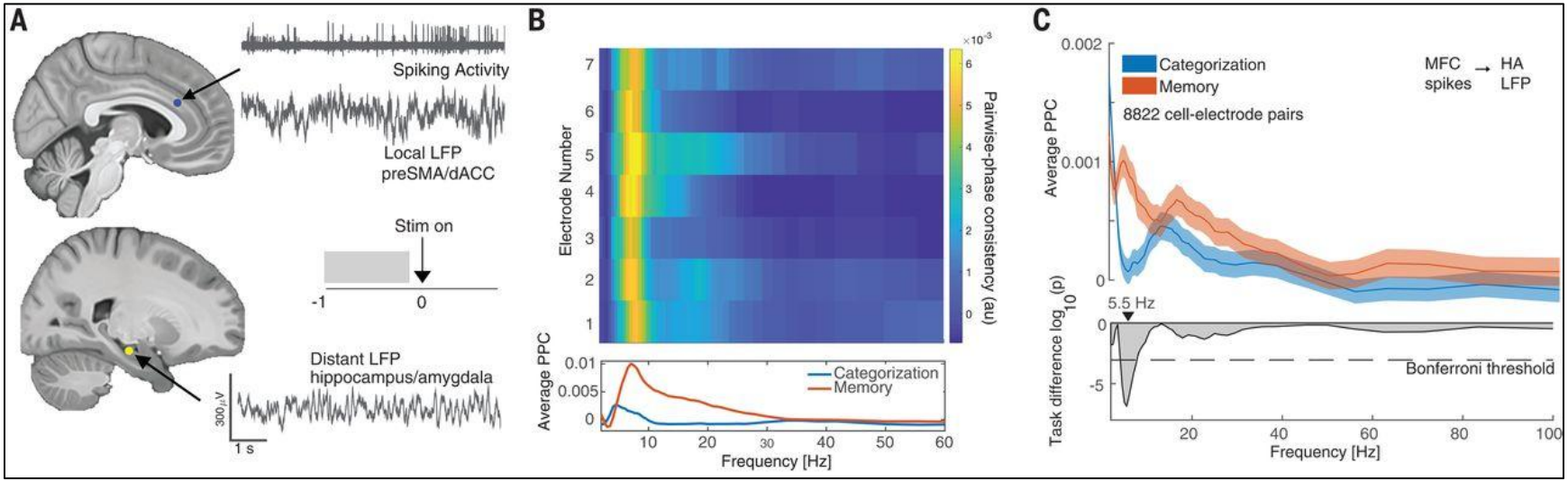
❖ Gaussian-process factor analysis (GPFA).

- Determined how individual cells are recruited by a linear decoder.
- (**K**) - Quantified its importance for both the memory and categorization choice decoder for each cell.

❖ Cells that report choice independently of task should lie on the diagonal.

➔ The distribution of angles was significantly bimodal across all cells, with modes centered away from the diagonal.

5. Task-dependent spike-field coherence btw MFC & HA

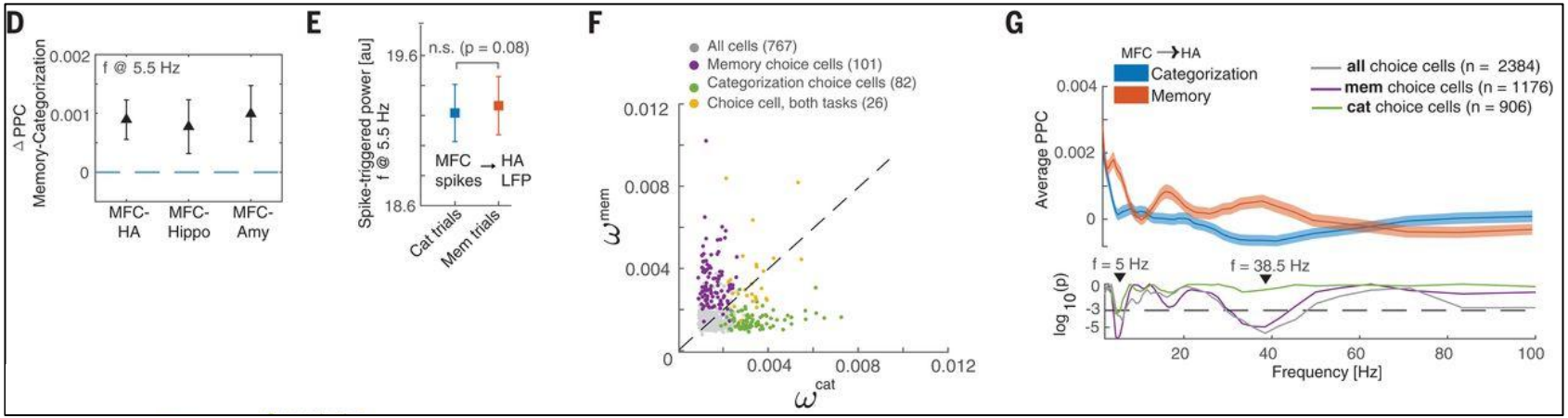


- It is thought that the selective routing of decision-related information between the HA and MFC is coordinated by interareal spike-field coherence. Therefore, asked whether MFC neurons phase-lock to ongoing oscillations in the LFP in the HA and, if so, whether the strength of such interactions is modulated by task demands.
- (**B**) - Individual cells in the MFC showed strong task modulation of MFC to HA spike-field coherence.

❖ Pairwise-phase consistency: bias-free rhythmic synchronization of neuronal activity.

- (**C**) - At the population level, MFC cells showed significantly stronger **theta-band coherence** with HA oscillations during the memory task than during the categorization task.

5. Task-dependent spike-field coherence btw MFC & HA



- (**D**) - This was the case for both MFC-hippocampus and MFC-amygdala interactions.

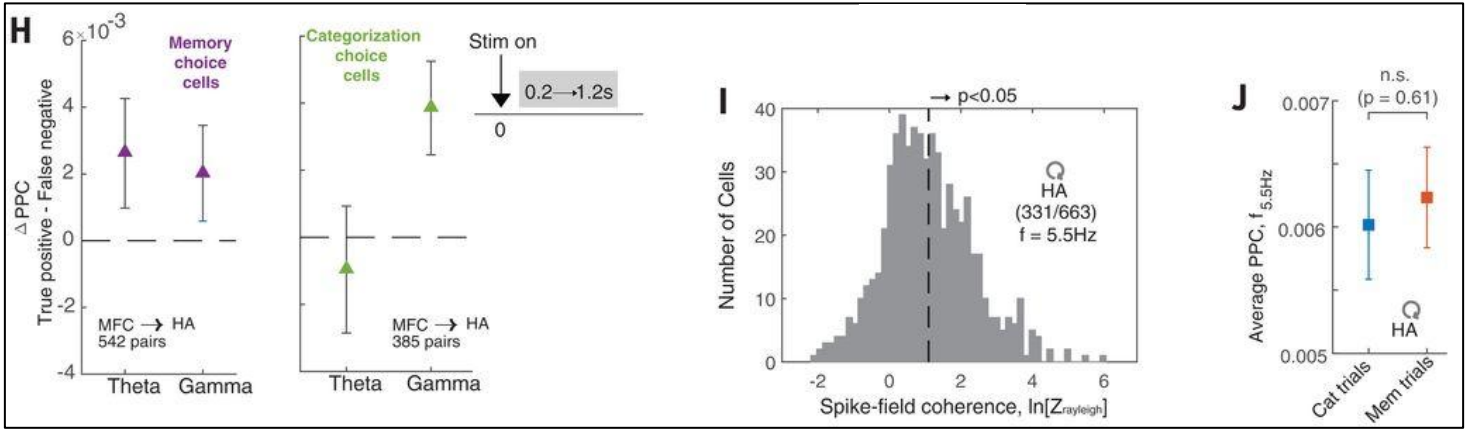
➔ (**E**) - This effect was due to changes in phase preference, as there was no significant difference in HA LFP power between the tasks.

- (**F**) - Each of the selected cells had a high importance index, as determined from population decoding.
- (**G**) - Considering only the MFC choice cells revealed that this subset of cells similarly increased their phase-locking during the memory task, with the strongest effect, again, in the theta range.

➔ Both categorization and memory choice cells showed this pattern of modulation.

➔ The memory choice cells exhibited an increase in gamma-band coherence.

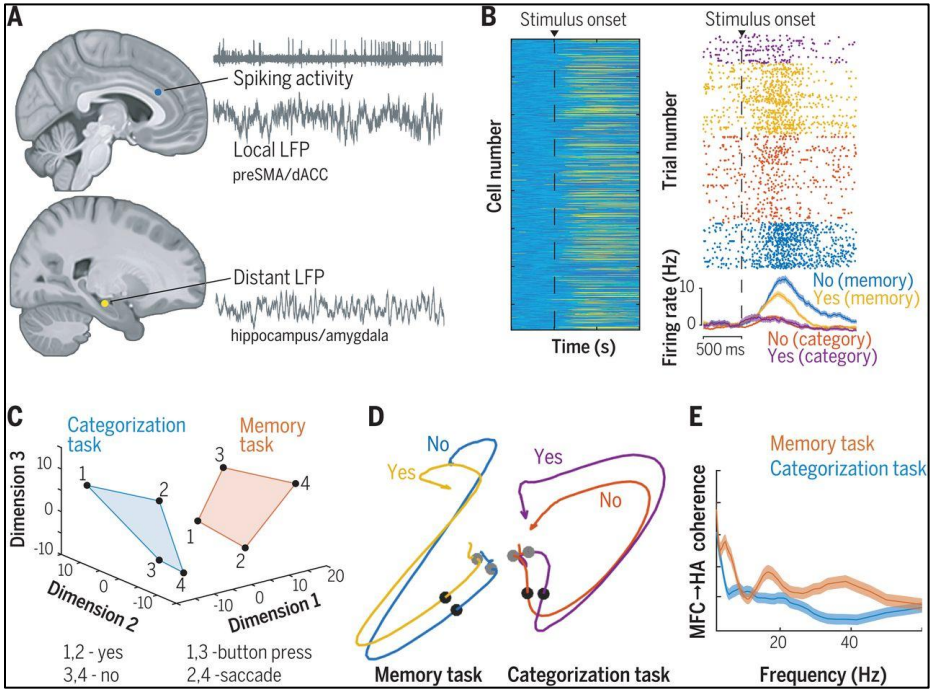
5. Task-dependent spike-field coherence btw MFC & HA



- (**H**) - The extent of phase-locking of choice cells following stimulus onset (0.2 to 1.2 s) during the memory task was significantly stronger for correctly retrieved trials than for forgotten old trials, indicating behavioral relevance for memory retrieval.
- (**I**) - The spiking activity of 331/663 HA cells was significantly related to the theta-frequency band LFP.
- (**J**) - The strength of this local spike-field coherence was, however, not significantly different between the two tasks.

Conclusion

1. Neuronal populations within the MFC formed two separate decision axes:
   - one for memory-based decisions and
   - another for categorization-based decisions.
2. MFC-HA theta-frequency functional connectivity was selectively enhanced during memory retrieval.
3. Oscillation-mediated coordination of activity between distant brain regions and accompanying changes in strength of representation and/or geometry implements task-dependent retrieval of memory.



References

1. Minxha et al., Science 368, eeba3313 (2020). DOI: 10.1126/science.aba3313